

Xiang He

Ph.D. Candidate in Geosciences, The University of Texas at Austin

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Research Profile

Computational geodynamicist studying how mantle flow, subduction, lithospheric deformation, and mantle rheology shape observable surface motions. I build physics-based numerical models and compare model predictions with geological and geophysical constraints, with current emphasis on Western U.S. vertical motion and Cascadia.

Education

The University of Texas at Austin

Ph.D. in Geosciences / Geophysics

Aug. 2024–Present

Austin, TX, USA

- Advisor: Prof. Thorsten W. Becker
- Research focus: computational geodynamics, mantle flow, subduction, dynamic topography, vertical crustal motion

University of Science and Technology of China

B.S. in Geophysics

Sep. 2020–Jun. 2024

Hefei, Anhui, China

- Guo Moruo Scholarship; China National Scholarship

Research Experience

Western U.S. Vertical Motion — UT Austin

Aug. 2024–Present

Graduate Researcher, UT Institute for Geophysics, Advisor: Thorsten Becker

Austin, TX

- Build global and regional geodynamic models to quantify how subduction, mantle flow, and rheology influence vertical crustal motion and dynamic topography.
- Develop Cascadia subduction-zone model configurations and compare model predictions with geological/geophysical observations.
- Design numerical experiments to test sensitivity to slab geometry, mantle rheology, density structure, and boundary conditions.
- Run and analyze large-scale simulations using ASPECT and high-performance computing workflows; process model output with Python, GMT, and related tools.

Western Pacific double-subduction drives Gulf of California rifting — UT Austin

Feb. 2025–Jul. 2026

Graduate Researcher, UT Institute for Geophysics, Advisor: Thorsten Becker

Austin, TX

- Set up global reference geodynamic model to evaluate velocity field and stress state of lithosphere.
- Compare double and single subduction configurations, evaluate the additional extensional stress (15 MPa) imparted on Gulf of California rifting from western Pacific double-subduction system.

3D Regional Model: Intraplate Deformation within East Asia (remote) — UIUC

May. 2023–Oct. 2023

Visiting Student, Department of Earth Science and Environmental Change, Advisor: Lijun Liu

Urbana, IL

- Conducted time-dependent 3-D regional geodynamic modeling of intraplate deformation in East Asia.
- Analyse the velocity field and stress field near Tan-Lu Fault and investigate how its strike-slip pattern changes through time.

Rock Dynamics Simulation of Planetary Crust Formation — USTC

Jan. 2023–Aug. 2023

Undergraduate Researcher, School of Earth and Space Sciences, Advisor: Wei Leng

Hefei, China

- Set up a 3D regional model to evaluate the effect of hydrated plume on continental crust formation.
- Model shows that the generation of continental crust is almost impossible with a dry plume, which indicates the necessity of deep water's participation in this process.

Réunion plume and double-subduction coupling as a driver of India's two-stage rapid northward motion — USTC

Apr. 2022–Jun. 2024

Undergraduate Researcher, School of Earth and Space Sciences, Advisor: Jinshui Huang

Hefei, China

- Develop time-dependent geodynamic models coupling early Réunion plume-lithosphere interaction (70-66 Ma) with an evolving double-subduction system, and quantify their respective contributions to India's acceleration, sustained rapid motion, and eventual slowdown.

Publications

- [1] G. Gianni, X. He, T. W. Becker, and C. Faccenna. "Western Pacific double-subduction drives Gulf of California rifting". In: *Science* (2026). Accepted.

Selected Research Themes

- **Western U.S. vertical motion and Cascadia:** subduction-system dynamics, mantle flow, dynamic topography, and comparison with surface observations.
- **Mantle rheology and plate motion:** sensitivity of plate-scale deformation and surface observables to mantle viscosity structure and density heterogeneity.
- **Subduction and plume dynamics:** numerical models of single/double subduction, plume–plate interactions, and long-term tectonic evolution.

Technical Skills

Programming: Python, C, Fortran, MATLAB, shell scripting

Geodynamic / Geophysical Software: ASPECT, CitcomS, GPlates, GMT / PyGMT, ParaView

Scientific Computing: HPC workflows, MPI-based simulations, Linux, Git, numerical modeling, inverse problems, data–model comparison

Methods: Finite-element geodynamics, sensitivity analysis, geophysical data processing, visualization

Awards & Honors

Graduate Fellowship, UT Austin	2024–2025
Guo Moruo Scholarship, USTC	2024
China National Scholarship (top 0.2% in China)	2023
Yang Yongman Scholarship, USTC	2023
USTC Outstanding Student Scholarship, Gold Medal (top 3%)	2022

Teaching

University of Science and Technology of China

Teaching Assistant, Electromagnetism A (PHYS1004A.04.2023SP)

Instructor: Prof. Gan Qin

Mar. 2023–Jun. 2023

Hefei, China